Firewall Configuration & Testing

Objective:

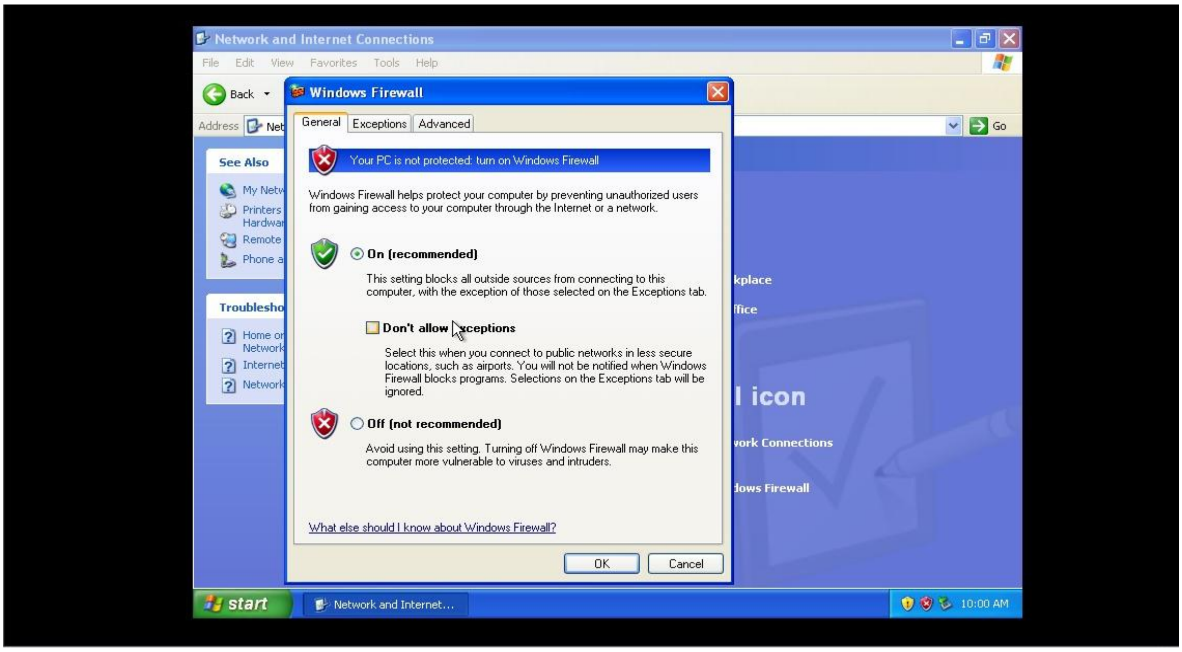
This task was performed to understand firewall rule configuration, how to block ports and IP addresses? How to read firewall logs? I configured firewall in Windows XP and used Kali Linux for scan the Windows XP.

Tools:

Firewall: Windows XP Firewall

Testing OS: Kali Linux

Scanning Tool: Nmap

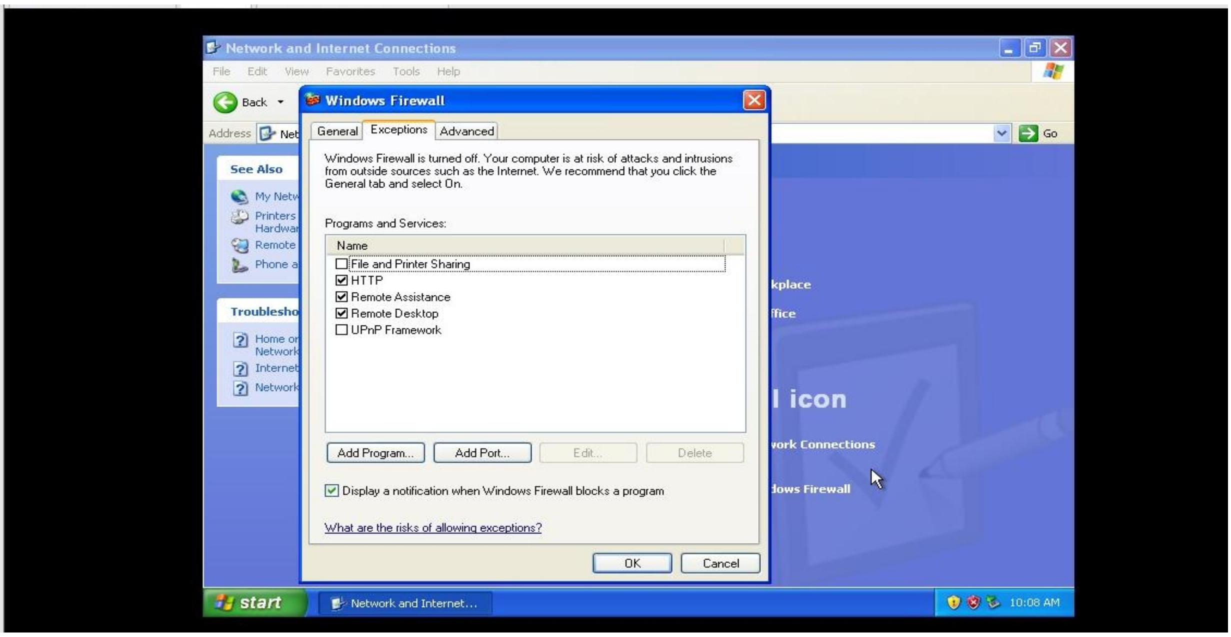


Firewall Configuration:

The following firewall rules were configured on the Windows XP system:

Service	Port	Firewall Rule

HTTP	80	Allowed
FTP	21	Filtered
Telnet	23	Filtered
RDP	3389	Allowed



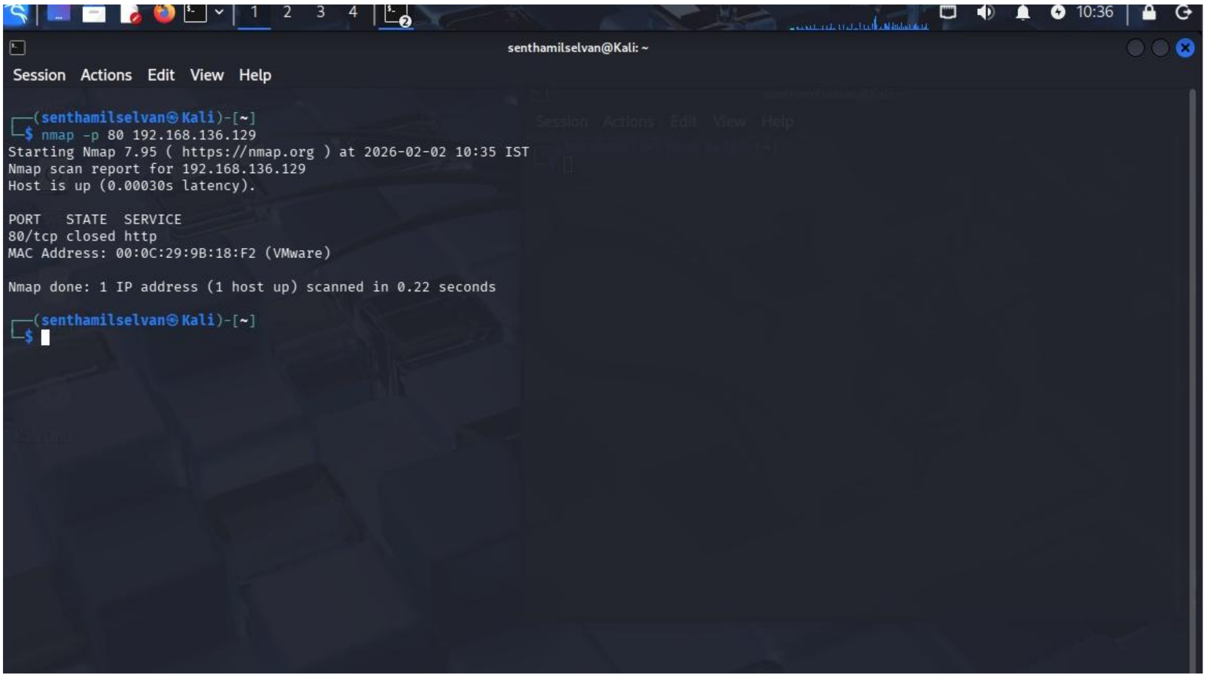
<u>Nmap Scan Results:</u>

Port	**Service**	**Status**	**Reason**
80	HTTP	Closed	No service running
21	FTP	Filtered	Blocked by firewall
23	Telnet	Filtered	Blocked by firewall
3389	RDP	Open	Service running

Results:

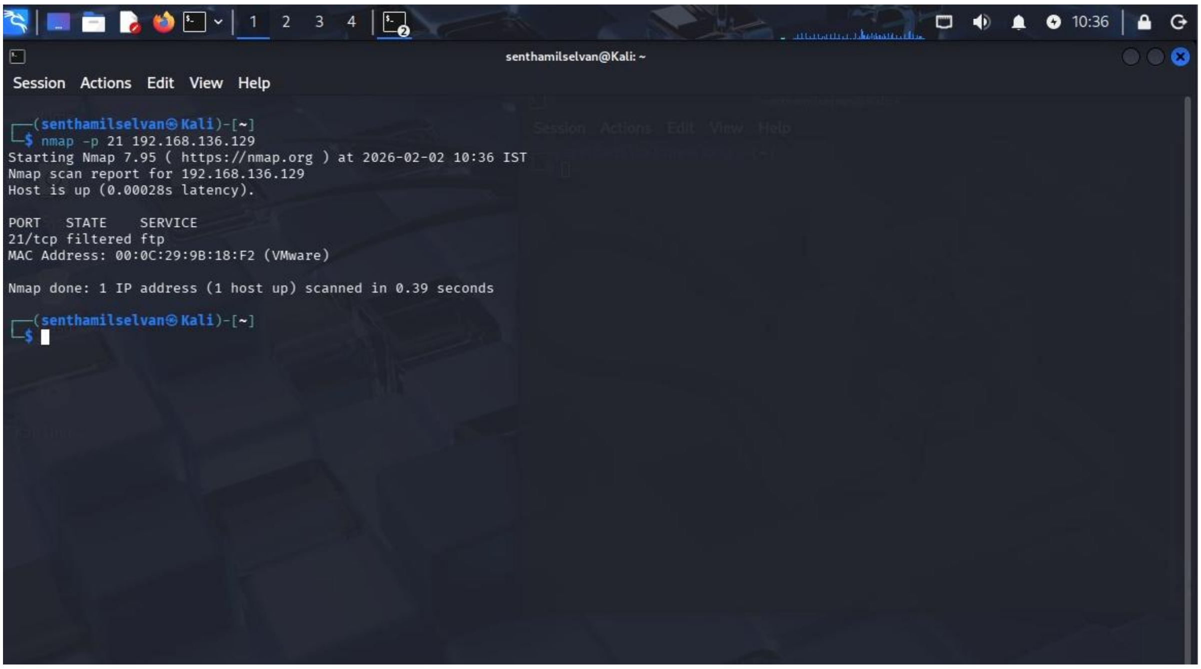
HTTP (Port 80 – Closed):

Although port 80 was allowed in the firewall, no web service was running on the system. Therefore, the port appeared as *closed* during Nmap scanning.



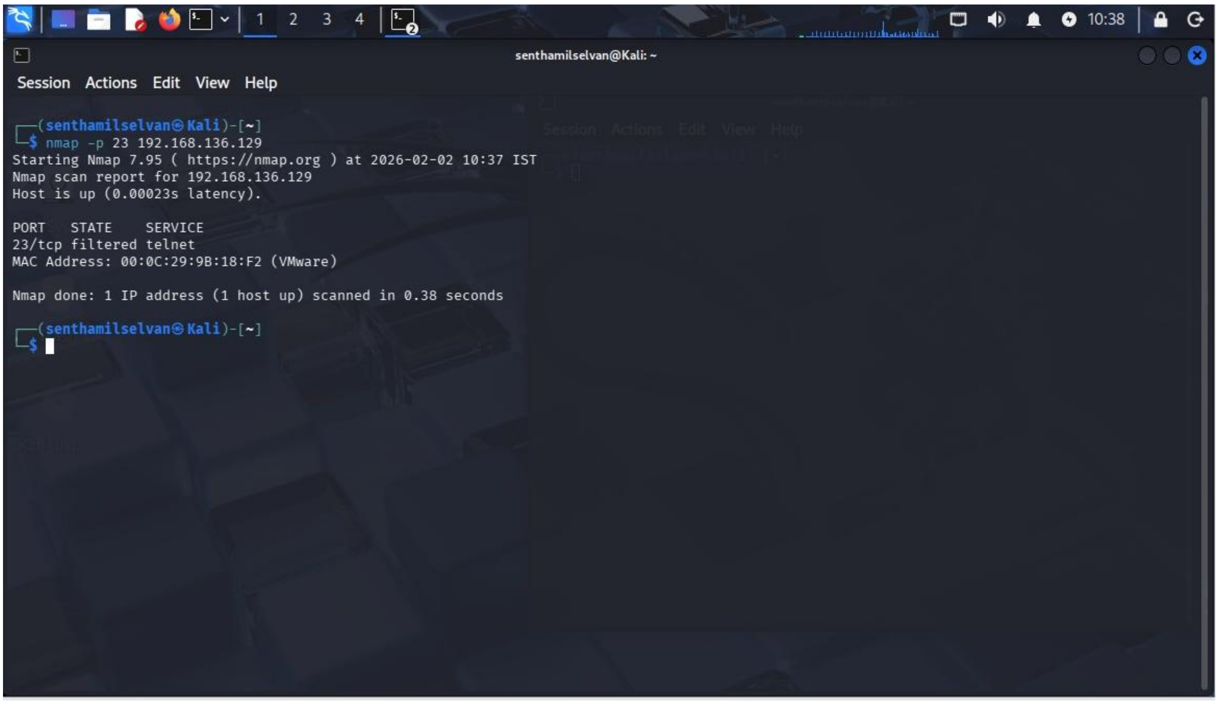
FTP (Port 21 – Filtered):

The firewall filtered FTP traffic, causing the port to drop packets silently and remain hidden from the scanner.



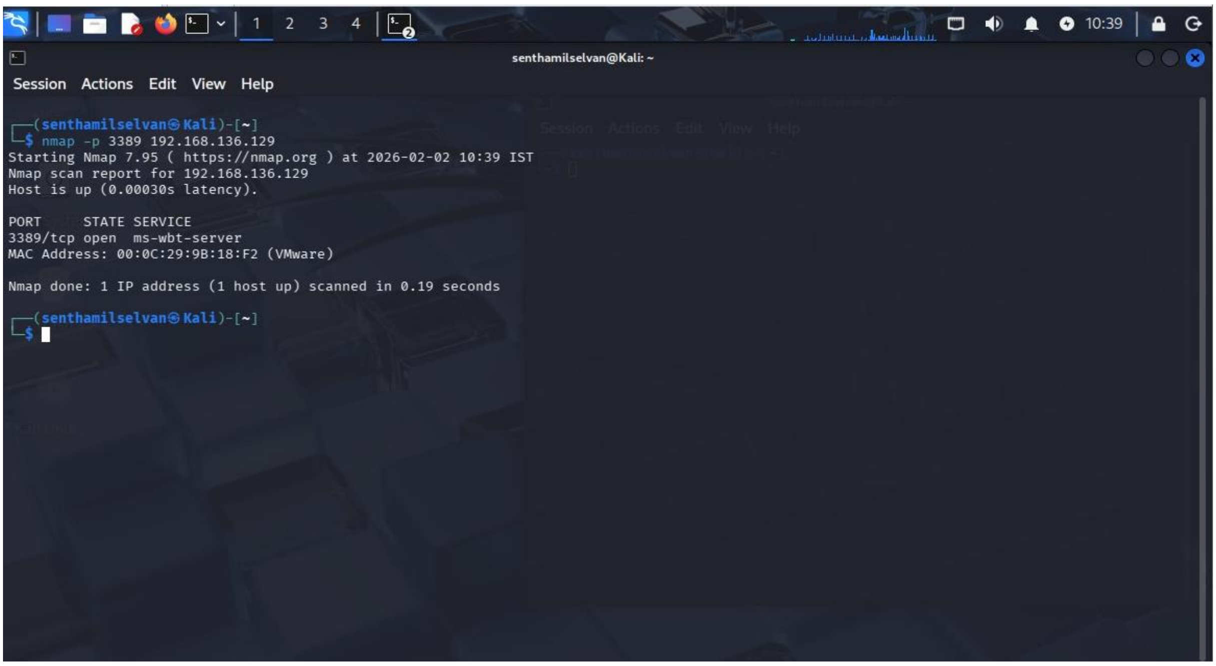
Telnet (Port 23 – Filtered):

Telnet traffic was filtered to prevent insecure remote access.



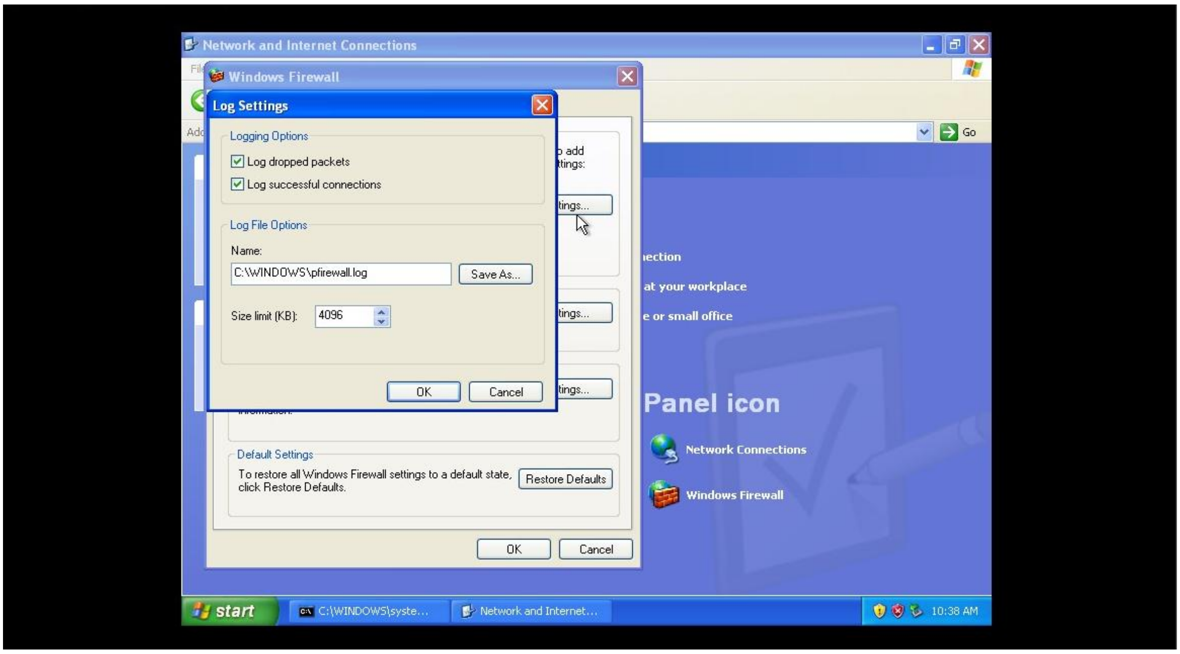
RDP (Port 3389 – Open):

The RDP service was running and allowed through the firewall, so the port appeared open.



LOG File:

I stored the XP firewall's log and analysed it. The Windows Firewall effectively blocked NetBIOS broadcasts and unauthorized TCP connection attempts to FTP and Telnet ports while allowing necessary services such as DHCP and controlled RDP access, indicating correct firewall configuration and successful mitigation of network scanning activity.



Key Learning:

- Firewall rules only control traffic; they do not start services.
- An allowed port without a running service will still show as closed.
- Filtered ports provide better security by hiding services.
- Port status depends on both firewall rules and service availability.

Impact of Configuration

- Reduced system exposure by filtering unnecessary services.
- Allowed required remote access securely.
- Improved understanding of firewall vs service behaviour.